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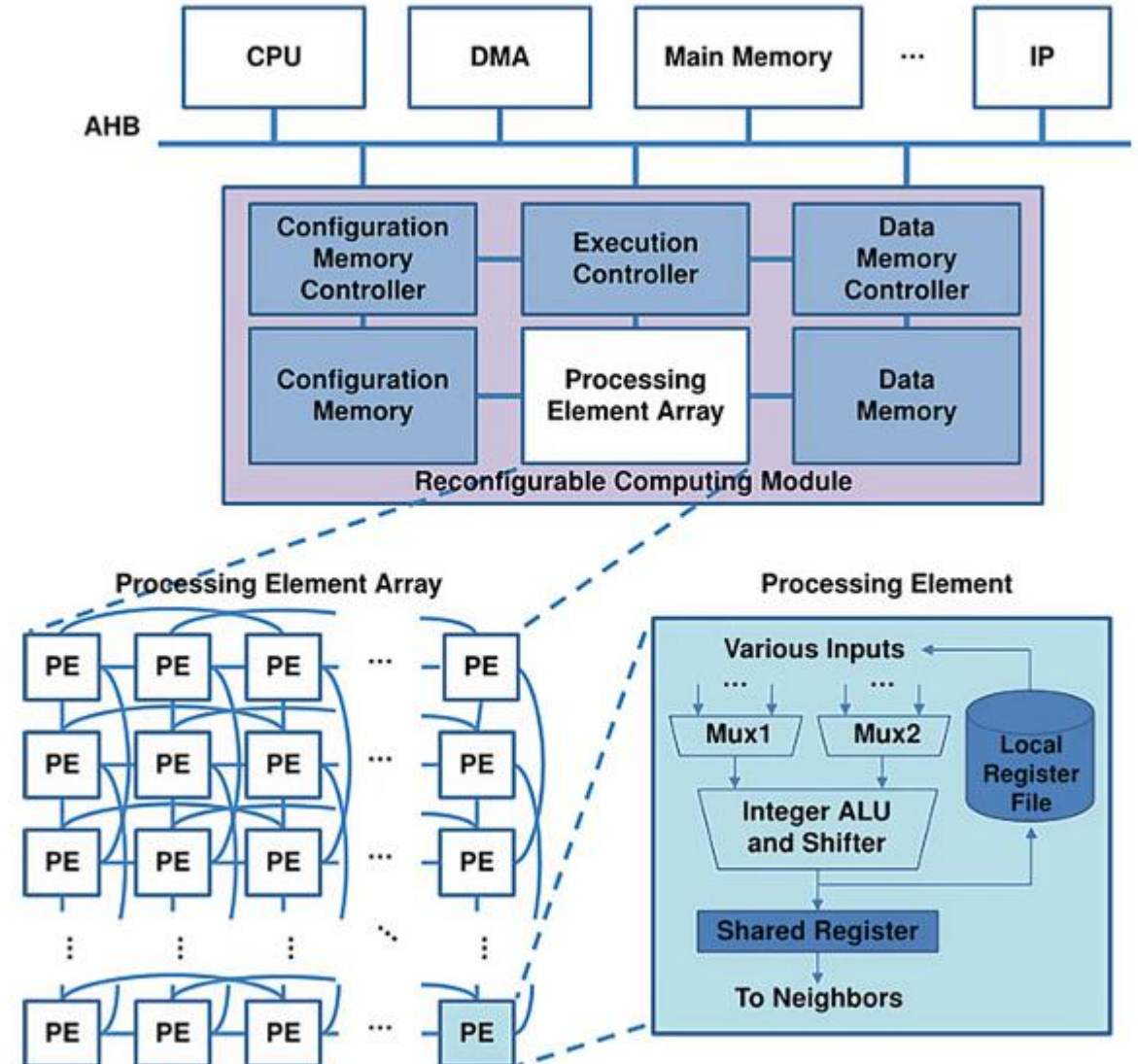


Embedded System Design
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Reconfigurable Architectures

Course Grained Reconfigurable Architecture

- The main part of CGRA is an array of PEs, RFs, and their interconnections.
- The array is connected to data and configuration memories as well as a host processor.
- A PE is basically a unit that performs ALU operations. A PE has its own registers to save temporary data.
- Host processor controls the reconfiguration of the array.



Execution controller

- **“micro-scheduler + state machine”** that ensures:
 - when each PE executes
 - how long each context is active
 - when to load the next configuration
 - when to start/stop pipelines
 - when to read/write memory
 - when to synchronize between PEs

FGRA Vs CGRA

- FPGA Programming = Logic-Level Programming
Hardware construction
- CGRA Programming = Operation-Level Scheduling (not logic-level)

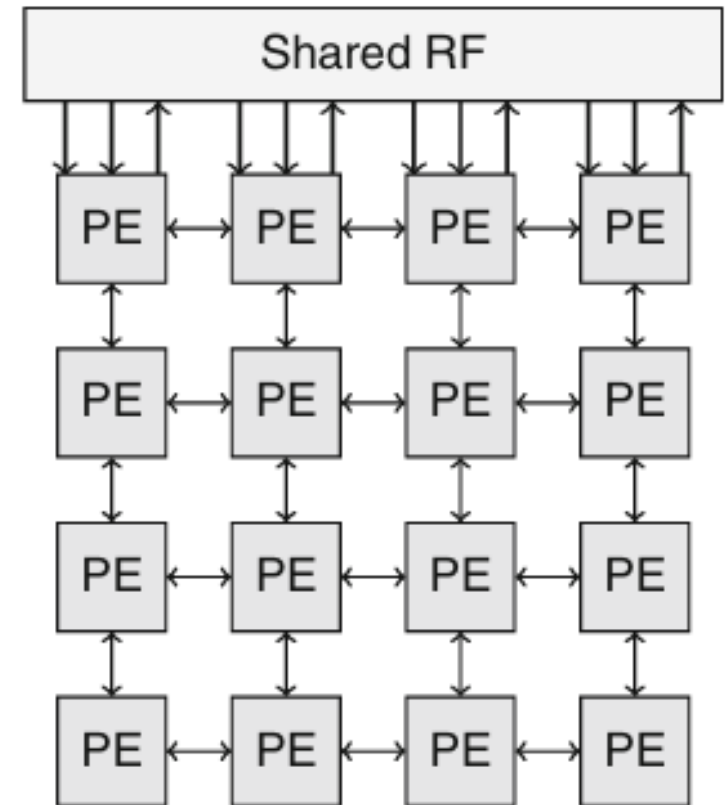
CGRA

- telling each PE what operation to execute (ADD, MUL, MAC, SHIFT...)
- specifying when to execute it
- specifying from where each PE gets its input
- specifying to whom it sends its output
- configuring routes between PEs

Nearest Neighbor (NN) Interconnect

- Each PE connects only to **4 neighbors**: left, right, up, down.
- Shared RF connected on top.
- Perfect for **local operations**
 - FIR filters
 - Matrix multiplications
- **Disadvantages**
 - Difficult to route non-local connections

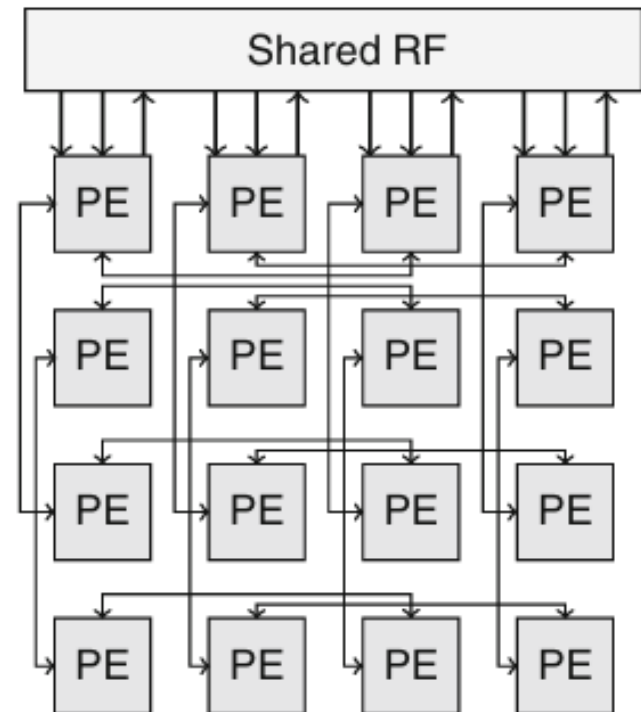
Eg: TI DSPs



Next-Hop (Flexible Switchbox) Interconnect

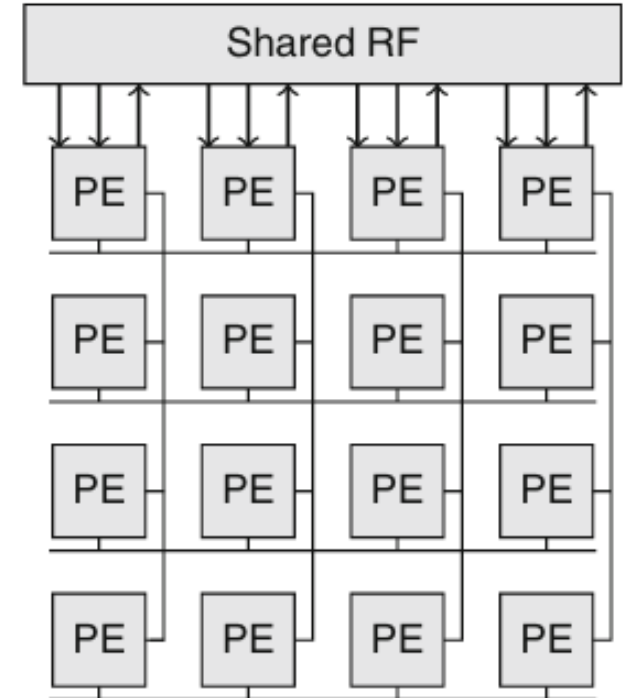
- Manhattan routing
- Each PE has a switch that can forward data multiple hops.
- PE is connected to a small crossbar or 3–4-way switch.

Samsung's CGRA-like video processors
NVIDIA ML Accelerators (Ampere)



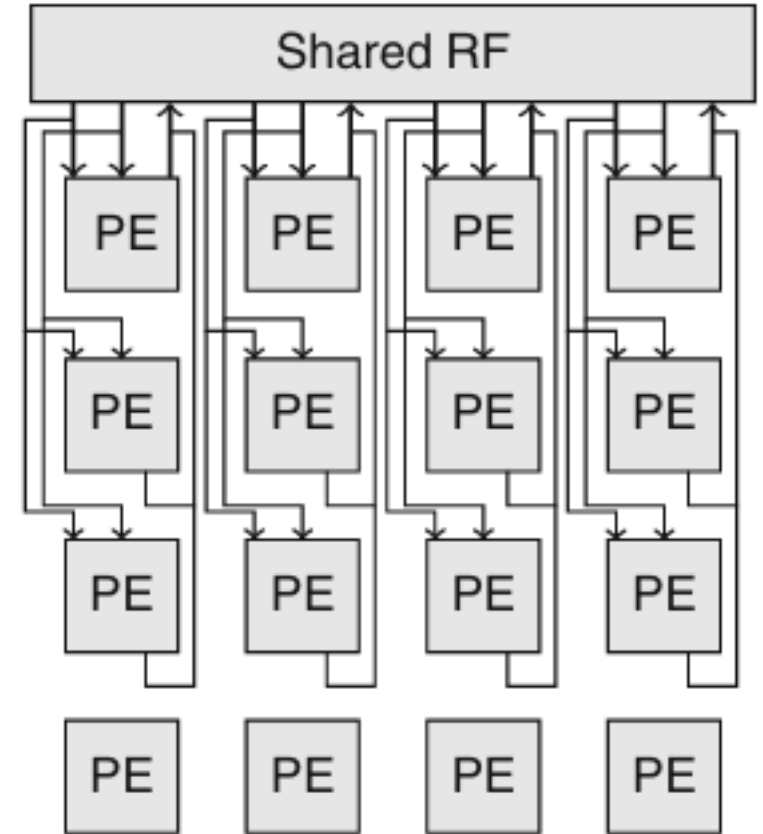
Bus-Based Interconnect

- Each PE can read from 2 bus
- Intel/Movidius Myriad- Intel Neural Compute Stick, DJI drones, Google Clips camera,
- Google Pixel Neural Core components



Extra (Augmented) Interconnect

- nearest neighbor + long-distance links + buses.
- Audio/video pipelines, ML accelerators



CGRA vs Microcontroller

- A CGRA contains:
 - Many distributed PEs (ALUs/MACs)
 - A reconfigurable interconnect
 - Context memory that defines operation *and* routing
 - The ability to build **custom dataflow circuits at runtime**

A CGRA program **reconfigures the hardware datapath**

Dataflow-style, instruction-less steady-state execution